Scientific Computing 2026

Samples, concentration, and limit theorems

Lecture 4

One hundred waiting times

Record 100 waits. Compute their average.
Repeat with another 100 waits.

$$T_i \sim \mathrm{Exp}(0.2), \qquad \mu = 5 \text{ min}, \qquad \sigma = 5 \text{ min}.$$

How far might either average be from five minutes?

Samples and sampling distributions

Definition: Sample mean and sampling distribution

An i.i.d. sample has mutually independent observations with the same distribution. Its sample mean is

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

The distribution of $\overline{X}_n$ across repeated datasets is its **sampling distribution**.

Before collection: random variables $X_1, \ldots, X_n$.
After collection: recorded values $x_1, \ldots, x_n$.

Standard deviation and standard error

For i.i.d. observations with finite variance,

$$\mathbb{E}[\overline{X}_n] = \mu, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

Definition: Standard error

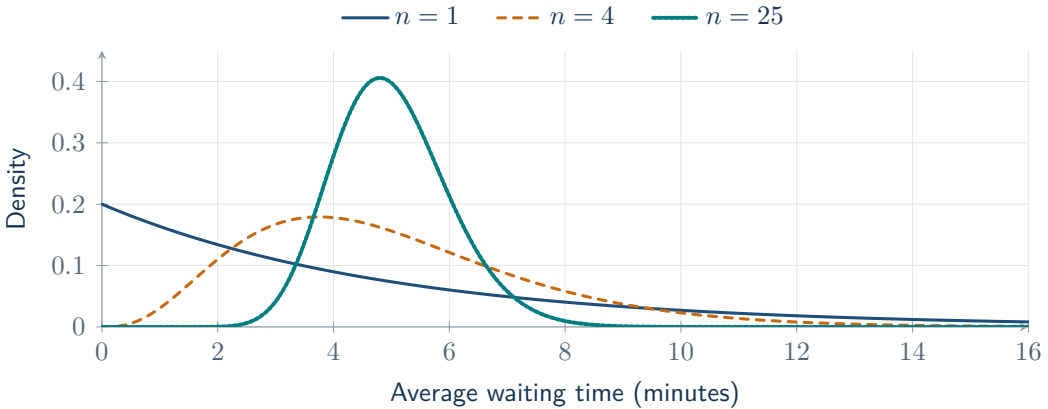
A statistic's standard deviation across repeated samples. For an i.i.d. sample mean,

$$\mathrm{SE}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

One wait: $\mathrm{SD} = 5$ minutes.
Average of 100 waits: $\mathrm{SE} = 0.5$ minutes.

Individual waits keep their spread as the sample grows.

The exact distribution of an average



$\overline{T}_n \sim \text{Gamma}(n, 0.2n)$, with rate $0.2n$. The mean stays at five minutes.

Unknown population variance

Definition: Sample variance

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2, \qquad n \geq 2.$$

The denominator accounts for centering at the sample mean. We derive the correction in the estimation chapter.

$$\widehat{\mathrm{SE}}(\overline{X}_n) = \frac{S_n}{\sqrt{n}}, \qquad S_n = \sqrt{S_n^2}.$$

Here the observations are i.i.d. with finite population variance.

The probability of a large error

Suppose the average wait should be within one minute of the population mean.

$$\mathbb{P}\big(\big|\overline{T}_{100} - 5\big| \geq 1\big) = ?$$

The standard error gives an error scale.
A probability statement needs more reasoning.

Markov's inequality

Theorem: Markov's inequality

If $W \geq 0$, $\mathbb{E}[W] < \infty$, and $a > 0$, then

$$\mathbb{P}(W \geq a) \leq \frac{\mathbb{E}[W]}{a}.$$

Proof. On the event $W \geq a$, the value of W is at least a:

$$W \geq a\,\mathbf{1}_{\{W \geq a\}} \quad \Longrightarrow \quad \mathbb{E}[W] \geq a\,\mathbb{P}(W \geq a).$$

Frequent large values force the mean upward.

Why Markov can be conservative

Any nonnegative wait with mean five minutes:

$$\mathbb{P}(T \geq 20) \leq \frac{5}{20} = 0.25.$$

Under the exponential model:

$$\mathbb{P}(T \geq 20) = e^{-4} \approx 0.0183.$$

The bound is attainable: probability $\frac{1}{4}$ at 20 and $\frac{3}{4}$ at 0. A mean alone cannot distinguish these distributions.

Chebyshev's inequality

Theorem: Chebyshev's inequality

If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then for $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov with $W = (X - \mu)^2$ and $a = \varepsilon^2$:

$$\mathbb{P}(|X - \mu| \geq \varepsilon) = \mathbb{P}(W \geq \varepsilon^2).$$

Squaring deviations makes Markov apply to both tails.

A finite-sample guarantee

$$\mathbb{P}(|\overline{X}_n - \mu| \geq \varepsilon) \leq \min\left(1, \frac{\sigma^2}{n\varepsilon^2}\right).$$

For the inbox, $\varepsilon = 1$ minute and $n = 100$:

$$\mathbb{P}(|\overline{T}_{100} - 5| \geq 1) \leq \frac{25}{100} = 0.25.$$

To make the bound at most δ, it suffices that

$$n \geq \frac{\sigma^2}{\delta\varepsilon^2}. \qquad \delta = 0.05 \text{ gives } n = 500.$$

Use the true variance or a known upper bound. A sample-variance plug-in needs further justification.

The weak law of large numbers

Theorem: Weak LLN, finite-variance version

For i.i.d. observations with mean μ and finite variance, every fixed tolerance $\varepsilon > 0$ satisfies

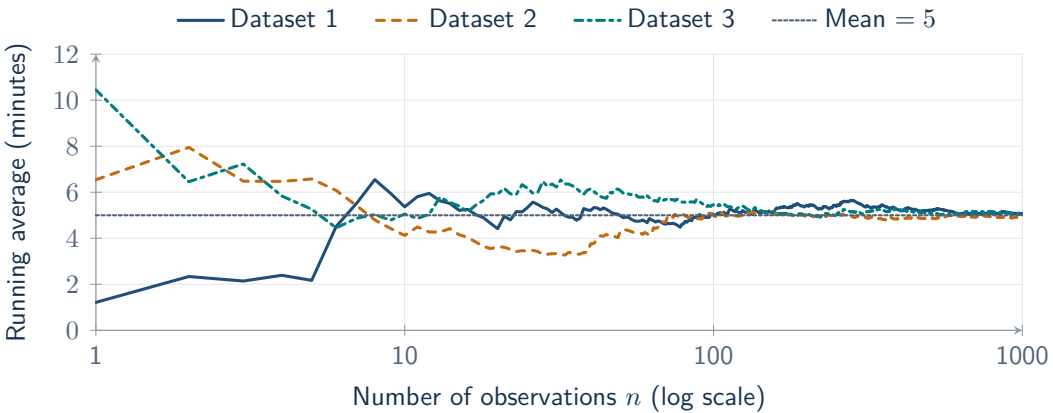
$$\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Proof. Chebyshev gives

$$\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

This is convergence in probability.

Running averages fluctuate



A new observation can increase the error. These paths illustrate the LLN.

The scope of the law of large numbers

The fixed-tolerance error probability tends to zero across repeated datasets.

Theorem: LLN with a finite first moment

For i.i.d. observations, $\mathbb{E}[|X_1|] < \infty$ is enough:

$$\overline{X}_n \xrightarrow{\mathbb{P}} \mathbb{E}[X_1].$$

Finite variance is enough for our short proof.

Sampling bias remains: recording only short waits learns the mean of that selected population.

The shape of the sampling error

The LLN makes the error small.
What distribution describes the remaining fluctuations?

$$\overline{X}_n - \mu \qquad \text{has standard deviation} \qquad \frac{\sigma}{\sqrt{n}}.$$

The exponential example has an exact Gamma answer.
For a complicated measurement or loss, we need an approximation.

Standardized sampling error

Definition: Standardized sample mean

For $\sigma > 0$, measure the error in standard-error units:

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sqrt{n}\,(\overline{X}_n - \mu)}{\sigma}.$$

$$\mathbb{E}[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.$$

Rescaling keeps the shrinking fluctuations visible.
The new claim concerns their distributional shape.

The central limit theorem

Theorem: Classical CLT

For i.i.d. observations with mean μ and $0 < \sigma^2 < \infty$,

$$\mathbb{P}(Z_n \leq z) \longrightarrow \Phi(z) \qquad \text{as } n \to \infty,$$

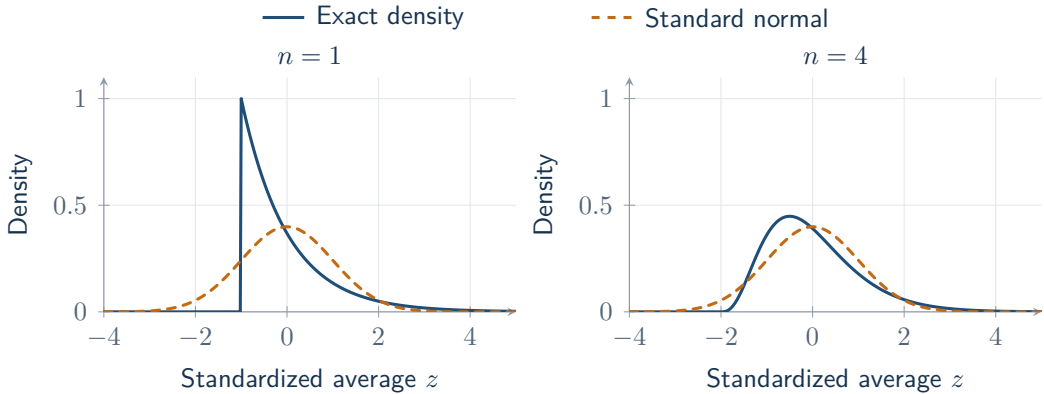
for every real z.

This is convergence in distribution to the standard Gaussian.

$$\overline{X}_n \overset{\text{approx.}}{\sim} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

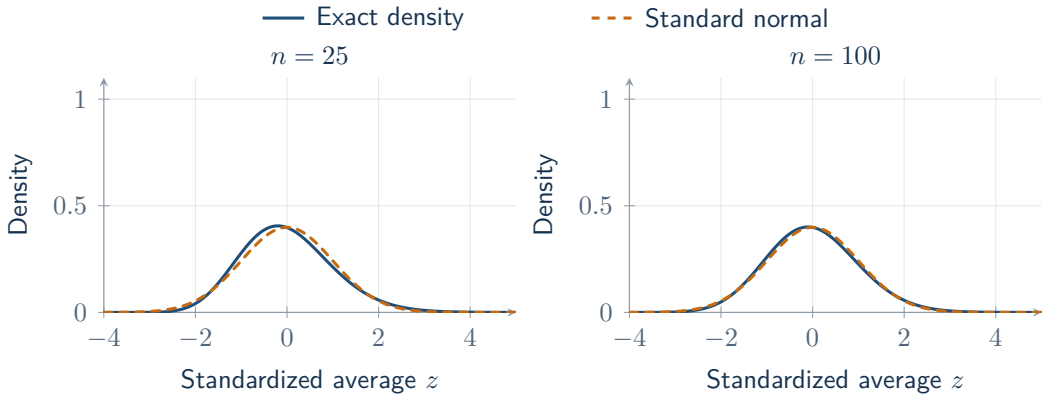
Gaussian observations give this law exactly for every n. The original observations keep their own distribution.

The CLT at small sample sizes



The support starts at $-\sqrt{n}$. Small-sample averages remain asymmetric.

The CLT at larger sample sizes



The same axes reveal the change in shape as n grows.

One minute equals two standard errors

$$n = 100, \qquad \mathrm{SE}(\overline{T}_{100}) = \frac{5}{\sqrt{100}} = 0.5.$$

$$\mathbb{P}(|\overline{T}_{100} - 5| > 1) = \mathbb{P}(|Z_{100}| > 2)$$

$$\approx 2\,[1 - \Phi(2)] \approx 0.04550.$$

About 4.55% under the Gaussian approximation.

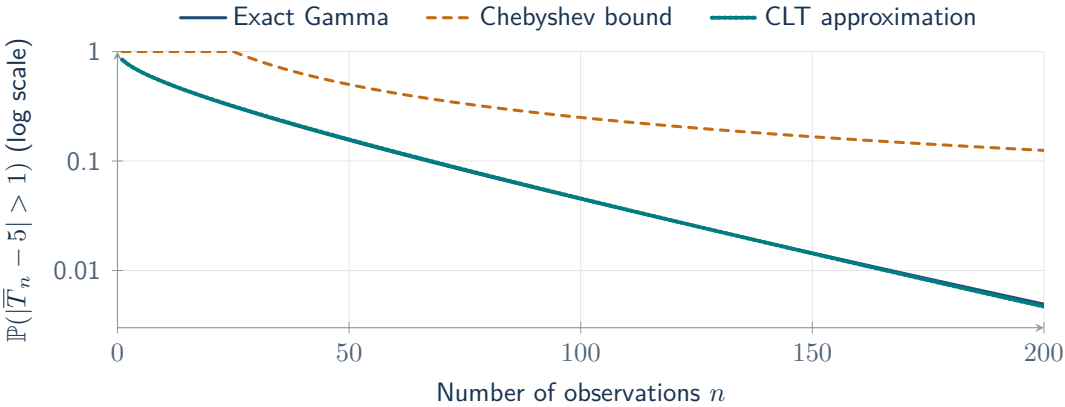
Bound, approximation, and exact probability

The Gamma model lets us check the approximation:

$$\mathbb{P}(\overline{T}_{100} < 4) \approx 0.017108,$$
$$\mathbb{P}(\overline{T}_{100} > 6) \approx 0.027864.$$

Method	Value	Meaning
Chebyshev	≤ 0.25000	Upper bound
CLT	≈ 0.04550	Approximation
Exact Gamma	≈ 0.04497	Exact model, rounded

Error probability across sample sizes



The exact and CLT curves nearly coincide here. Chebyshev stays an upper bound.

Finite-sample accuracy of the CLT

There is no universal sample size, such as 30, that guarantees a useful Gaussian approximation.

Skewness and rare extreme values can delay convergence.

A small absolute CDF error can still mean a large relative error for a tiny tail probability.

A shared calibration error

$$X_i = \mu + B + \epsilon_i.$$

B stays fixed within one dataset. A new independently calibrated dataset gets a fresh B.

The reading errors ϵ_i are centered and independent, and independent of the centered calibration error B.

$$\overline{X}_n = \mu + B + \frac{1}{n}\sum_{i=1}^{n} \epsilon_i.$$

Covariance leaves a variance floor

Each reading has variance σ^2. Distinct pairs have correlation ρ, with $0 \le \rho \le 1$.

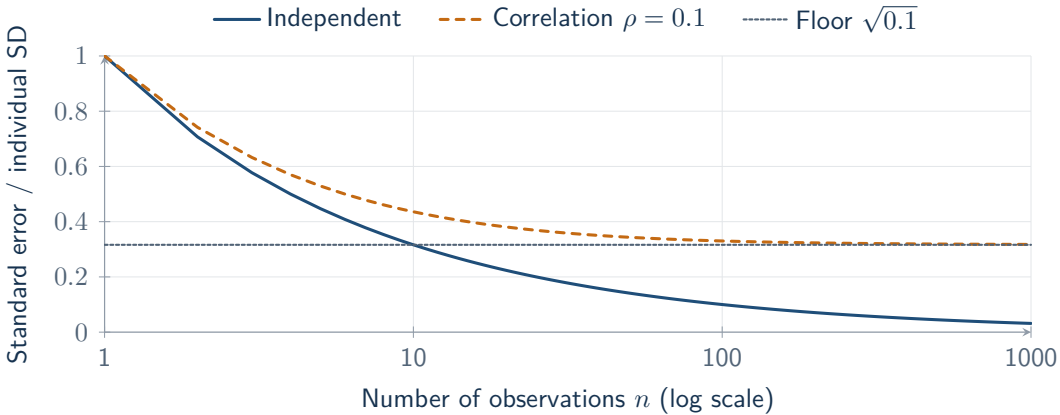
$$\begin{aligned}\operatorname{Var}(\overline{X}_n) &= \frac{n\sigma^2 + n(n-1)\rho\sigma^2}{n^2} \\ &= \sigma^2\left[\rho + \frac{1-\rho}{n}\right].\end{aligned}$$

There are n diagonal covariance terms and $n(n-1)$ off-diagonal terms.

Key result: Variance floor

$$\operatorname{Var}(\overline{X}_n) \longrightarrow \rho\sigma^2 \qquad \text{as } n \to \infty.$$

One hundred correlated readings



For $n = 100$, $\mathrm{Var}(\overline{X}_{100}) = 0.109\sigma^2$:
the same mean variance as about 9.17 independent readings.

Finite mean and finite variance

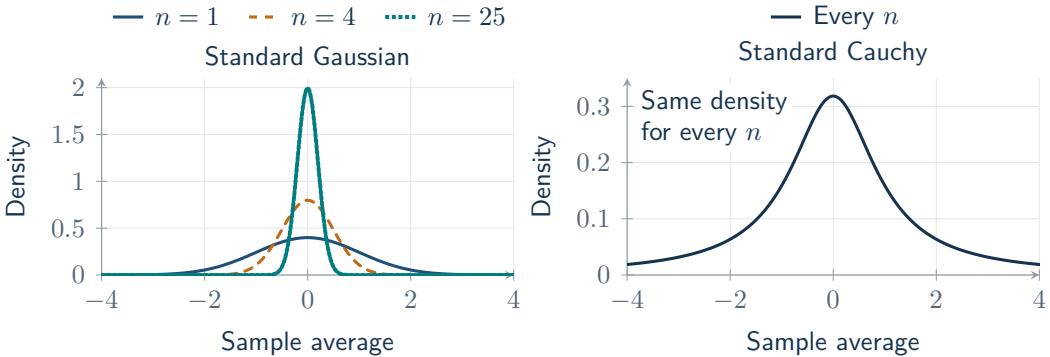
For i.i.d. Student observations with $\nu = \frac{3}{2}$:

$$\mathbb{E}[|X_1|] < \infty, \qquad \mathrm{Var}(X_1) = \infty.$$

The general LLN applies: the sample mean converges in probability to zero.

The finite-variance Chebyshev argument and classical CLT do not apply. There is no finite σ for $\sigma/\sqrt{n}$.

Averages of i.i.d. standard Cauchy observations



$$\overline{X}_n \sim t_1, \qquad \mathbb{P}(|\overline{X}_n| > 1) = \tfrac{1}{2} \quad \text{for every } n.$$

The Cauchy mean is undefined. Zero is its symmetry center. Panel scales differ.

Monte Carlo estimation

Definition: Monte Carlo estimator

For $I = \mathbb{E}[g(X)]$, draw X_i independently from the target distribution:

$$\widehat{I}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i).$$

$\mathbb{E}[|g(X)|] < \infty$ gives the LLN.
Finite positive τ^2 also gives the CLT.

$$\mathrm{SE}(\widehat{I}_n) = \frac{\tau}{\sqrt{n}}, \qquad \tau^2 = \mathrm{Var}(g(X)).$$

For unknown τ: $\widehat{\mathrm{SE}} = S_Y/\sqrt{n}$, with S_Y the sample SD of $g(X_i)$.

A Monte Carlo example with a known answer

$$T \sim \mathrm{Exp}(0.2), \qquad g(t) = e^{-t}.$$

Time is in minutes. The exponent uses one per minute.

$$I = \mathbb{E}[g(T)] = \frac{1}{6}, \qquad \mathbb{E}[g(T)^2] = \frac{1}{11}.$$

$$\tau^2 = \frac{1}{11} - \frac{1}{36} = \frac{25}{396}$$

Precision has a computational cost

Definition: Root mean squared error

$$\mathrm{RMSE}(\widehat{I}_n) = \sqrt{\mathbb{E}[(\widehat{I}_n - I)^2]}.$$

For our Monte Carlo average, $\mathrm{RMSE}(\widehat{I}_n) = \tau/\sqrt{n}$.

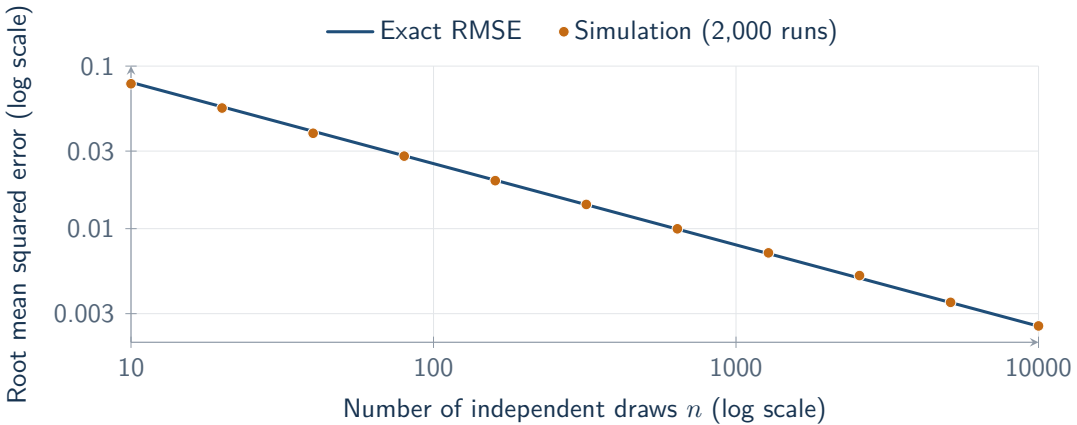
Four times as many evaluations halves RMSE.
A tenfold reduction costs one hundred times as many.

$$n = 1000 \qquad \Longrightarrow \qquad \mathrm{RMSE} \approx 0.00795.$$

$$\mathrm{RMSE} \leq 0.005 \qquad \text{needs } n \geq 2526.$$

An RMSE target does not bound the error in every run.

Monte Carlo error across repeated runs



The dots measure variation across independent runs, rather than one running path.

Uncertainty in a test-set average

Fix f before evaluation. Use i.i.d. test observations independent of all data used to choose f.

Definition: Risk and empirical risk

$$L_i = \ell(f(X_i), Y_i), \qquad R(f) = \mathbb{E}[L_i], \qquad \widehat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} L_i.$$

Population accuracy 0.8, with 400 independent test examples:

$$\mathrm{SE} = \sqrt{\frac{0.8 \cdot 0.2}{400}} = 0.02.$$

Test-sample variation only. Choosing f on this test set breaks the fixed-model premise.

Batch size and the variance of an average

Key result: Variance of a batch average

At a fixed model, average b independent losses, each with variance τ_L^2.

$$\mathrm{Var}(\text{batch average}) = \frac{\tau_L^2}{b}.$$

A fourfold increase in b halves the standard deviation.

For a fixed dataset, sampling indices with replacement uses its empirical loss variance. Dependence and sampling without replacement require different variance calculations.

What each result tells us

Result	Question it answers
Chebyshev	How large can a finite-sample error probability be?
Law of large numbers	Does the average approach the population quantity?
Central limit theorem	What shape do the rescaled fluctuations approach?

Independence, moment assumptions, and the sampling process determine which conclusion is justified.